A Common Recovery Transition for Classical and Quantum Information in Random Codes

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Reading a classical message requires less than restoring quantum coherence. For the same random encoder under Pauli noise, we prove that optimal complete-message decoding success and entanglement fidelity share a critical curve. An information-spectrum bracket holds for every exact unitary 2-design. With independent identical errors, fluctuations of the error self-information yield a common Gaussian crossover at positive rate, in mean and with high probability. For Clifford codes, inverse encoding, basis measurement, and classical decoding attain the classical optimum.

A quantum code must preserve logical populations and coherences. A message encoded in a logical basis state is insensitive to logical phase errors, which can alter a superposition. This distinction suggests that classical messages could remain decodable when quantum recovery fails. Does it produce a separate decoding threshold for a random quantum code?

We compare the two tasks for the *same* random encoder followed by Pauli noise. The classical receiver guesses the entire uniformly distributed logical-basis message; the quantum receiver restores entanglement with a reference. Both know the realized encoder and optimize their decoders separately. At positive rate, their optimal performances approach the same Gaussian curve across the coding transition. The classical task therefore has no separate threshold or second-order rate advantage in this ensemble, despite its finite-size advantage. This is a statement about complete-message success and entanglement fidelity; partial information about the message can remain when complete decoding is unlikely.

The hashing rate $1 - H(\boldsymbol{p})$ is an established quantum-coding benchmark [1, 2]. Klesse's random-code argument already yields truncated-error-spectrum quantum achievability, including a probability bound over encoders [3]. Second-order quantum achievability [4] and finite-size Clifford analyses [5] quantify attainable rates. Haar-code spectra and coherent information exhibit a hashing transition with a square-root finite-size window [6]; approximate quantum Hamming bounds [7] and stabilizer converses [8] provide complementary constraints. Here a classical decoding converse matches the quantum achievable profile throughout the critical window. Together they determine the optimal classical *and* quantum performance of the same sampled code. The finite-size bracket and its probability bound require only unitary second moments.

One encoder, two tasks.—Let $d = 2^n$ and $M = 2^k \le d$, with integer $k \ge 1$. Embed the logical space by $V_0|j\rangle = |j\rangle|0\rangle^{\otimes(n-k)}$ and encode with $V = UV_0$, where U is drawn from an exact unitary 2-design. Haar and uniform Clifford encoders are examples [9]. The same U encodes all messages; the codewords are orthogonal, not independent random states. A fixed Pauli channel acts after encoding:

$$\mathcal{N}_p(\rho) = \sum_{P\in\{I,X,Y,Z\}^{\otimes n}} p_P P\rho P^\dagger, \quad s(P) = -\log_2 p_P. \tag{1}$$

The distribution is independent of U and may include correlated errors. Define $\mathcal{F}(t) = \mathbb{P}_{P\sim p}[s(P) < t]$. All logarithms are base two.

For $\rho_j = \mathcal{N}_p(V|j\rangle\langle j|V^\dagger)$, the optimal classical success probability is

$$P_{\mathrm{c}}^*(V) = \max_{\{E_j\}} \frac{1}{M}\sum_j \mathrm{tr}(E_j\rho_j). \tag{2}$$

Let $|\Phi_M\rangle = M^{-1/2}\sum_j |j\rangle|j\rangle$ and $\mathcal{V}(\rho) = V\rho V^\dagger$. The optimal entanglement fidelity is

$$F_{\mathrm{q}}^*(V) = \max_{\mathcal{D}} \langle\Phi_M|[\mathrm{id}\otimes(\mathcal{D}\mathcal{N}_p\mathcal{V})](\Phi_M)|\Phi_M\rangle, \tag{3}$$

where $\mathcal{D}$ is any quantum channel back to the logical space. Fidelity here is the overlap, not its square root. Both optimizations precede averaging over U. Encoding and decoding are ideal; neither a noise record nor access to the environment is supplied.

Bit errors and phase errors.—Clifford encoders make the distinction concrete. Conjugating a physical Pauli through the known inverse $U^\dagger$ gives logical bit-flip and phase labels (x, z) and an auxiliary bit-flip syndrome s. Let $w(s, x, z)$ sum the probabilities of all errors with those labels, including auxiliary phase degeneracy. The orthogonal syndrome sectors give [8]

$$\begin{aligned} F_{\mathrm{q}}^* &= \sum_s \max_{x,z} w(s,x,z), \\ P_{\mathrm{c}}^* &= \sum_s \max_x \sum_z w(s,x,z). \end{aligned} \tag{4}$$

Quantum recovery corrects the most likely logical Pauli in each sector; classical decoding can add the probabilities of all phase labels for the same bit flip. This is its finite-size advantage. The classical output states commute, so inverse encoding, basis measurement, and classical maximum-likelihood processing attain P_{c}^*. The processing can be computationally costly. The Supplemental Material (SM) [10] proves both optimality statements.

We now compare the tasks for any exact unitary 2-design, including non-Clifford encoders.

A common finite-size bracket.—The error spectrum constrains both optimized tasks.

Theorem 1. *For any distribution in Eq. (1), any exact unitary 2-design encoder, and every $a > 0$,*

$$\mathcal{F}(n-k-a) - 8\,2^{-a} \le \mathbb{E}F_{\mathrm{q}}^{*} \le \mathbb{E}P_{\mathrm{c}}^{*} \le \mathcal{F}(n-k+2a) + \frac{5}{4}\,2^{-a} + \frac{2^{a}}{4M}. \tag{5}$$

Moreover, $F_{\mathrm{q}}^{*}(V) \le P_{\mathrm{c}}^{*}(V)$ for every encoder. With $F_{-} = \mathcal{F}(n-k-a)$ and $F_{+} = \mathcal{F}(n-k+2a)$, for any $\eta > 0$,

$$\mathbb{P}_U[F_{\mathrm{q}}^{*} < F_{-} - \eta \text{ or } P_{\mathrm{c}}^{*} > F_{+} + \eta] \le \frac{37\,2^{-a}}{4\eta} + \frac{2^{a}}{4M\eta}. \tag{6}$$

The scale $n-k$ has a familiar coding interpretation. For i.i.d. errors with single-qubit Shannon entropy H, there are roughly 2^{nH} typical errors, each mapping a 2^k-dimensional code into the 2^n-dimensional physical space. Comparing $2^k 2^{nH}$ with 2^n suggests the hashing rate $R = k/n = 1 - H$. This counting picture motivates the scale; the theorem controls the actual overlaps and degeneracies without assuming orthogonal error images. Its converse shows that allowing classical decoding to combine phase errors does not shift the asymptotic complete-message threshold. At finite size, the full error spectrum, including its fluctuations, enters the bracket.

Proof.—For a recovered logical channel with Kraus operators A_ℓ,

$$F_e = \frac{1}{M^2}\sum_{\ell} |\operatorname{tr} A_\ell|^2 \le \frac{1}{M}\sum_{j,\ell} |(A_\ell)_{jj}|^2 \le P_{\mathrm{c}}^{*}. \tag{7}$$

The middle expression is the success of that recovery followed by a logical-basis measurement. This proves the pointwise ordering.

For the lower bound, take $\mathcal{G} = \{s < n-k-a\}$, $L = |\mathcal{G}| < d2^{-a}/M$, and define

$$S = \sum_{P\in\mathcal{G}} PVV^{\dagger}P^{\dagger}, \qquad R_P = V^{\dagger}P^{\dagger}S^{-1/2}. \tag{8}$$

The inverse is on the support of S. The R_P obey $\sum_P R_P^{\dagger}R_P = \Pi_{\mathrm{supp}S}$; completing them on its kernel gives a trace-preserving square-root recovery [11]. For $v_{jP} = PV|j\rangle$ and $a_{jP} = \langle v_{jP}|S^{-1/2}|v_{jP}\rangle$, Hayashi–Nagaoka [12] gives

$$1 - a_{jP}^2 \le 4 \sum_{(i,Q)\neq(j,P)} |\langle v_{jP}|v_{iQ}\rangle|^2 =: 4C_{jP}. \tag{9}$$

Also $0 \le a_{jP} \le 1$. Keeping the terms where the actual error matches the recovery Kraus index in the fidelity sum yields

$$F_e \ge \sum_{P\in\mathcal{G}} p_P \left(\frac{1}{M}\sum_j a_{jP}\right)^2 \ge p(\mathcal{G}) - 8X, \qquad X = \sum_{P\in\mathcal{G}} \frac{p_P}{M}\sum_j C_{jP}. \tag{10}$$

The label P in Eq. (8) is an internal Kraus index, not a known noise realization. Exact unitary second moments and orthogonality of the code columns give

$$\mathbb{E}C_{jP} = \frac{(L-1)(dM-1)}{d^2-1} \le \frac{ML}{d}, \tag{11}$$

so $\mathbb{E}X \le 2^{-a}$. This proves the lower bound, including nonuniform error probabilities. The empty $\mathcal{G}$ case is trivial.

For the upper bound, every POVM and every $c > 0$ satisfy

$$P_{\mathrm{succ}} \le \frac{cd}{M} + \frac{1}{M}\sum_j \operatorname{tr}(\rho_j - cI)_{+}. \tag{12}$$

Split each output at error probability c' into a head and a tail B_j. The head contributes at most its mass, while $(\lambda - c)_{+} \le \lambda^2/(4c)$ bounds the tail. Each code column is a state 2-design, and Pauli orthogonality gives

$$\mathbb{E}\operatorname{tr} B_j^2 = \frac{dv_t + m_t^2}{d+1} \le c' + \frac{1}{d}, \tag{13}$$

where $m_t = \sum_{p_P \le c'} p_P$ and $v_t = \sum_{p_P \le c'} p_P^2 \le c' m_t$. Putting $\gamma = cd$ and $c' = c2^{-a}$ gives

$$\mathbb{E}P_{\mathrm{c}}^{*} \le \frac{\gamma}{M} + \frac{1}{4\gamma} + \frac{2^{-a}}{4} + \mathcal{F}(n - \log_2\gamma + a). \tag{14}$$

Choose $\gamma = M2^{-a}$. The same pointwise constructions control the positive deviations from F_{-} and F_{+}; Markov's inequality gives Eq. (6). Full details appear in the SM [10].

Error fluctuations and the shared critical curve.—For independent identical errors, $p = \boldsymbol{p}^{\otimes n}$, write $H = \mathbb{E}[-\log_2 p_\alpha]$ and $\sigma^2 = \mathrm{Var}[-\log_2 p_\alpha]$ for one qubit. The total error self-information $s(P)$ is a sum of n independent terms, with mean nH and standard deviation $\sqrt{n}\,\sigma$. For $0 < H < 1$ and $\sigma > 0$, moving the coding budget $n-k$ through that distribution produces a window of width $O(\sqrt{n})$ in k, or $O(n^{-1/2})$ in rate. The Gaussian form comes from the central limit theorem for error self-information. Equation (5) transfers that form to both optimized recovery performances through matching quantum achievability and a classical converse.

At fixed positive rate $R = k/n$, the bracket gives exponential approach to one for $R < 1 - H$, and exponential

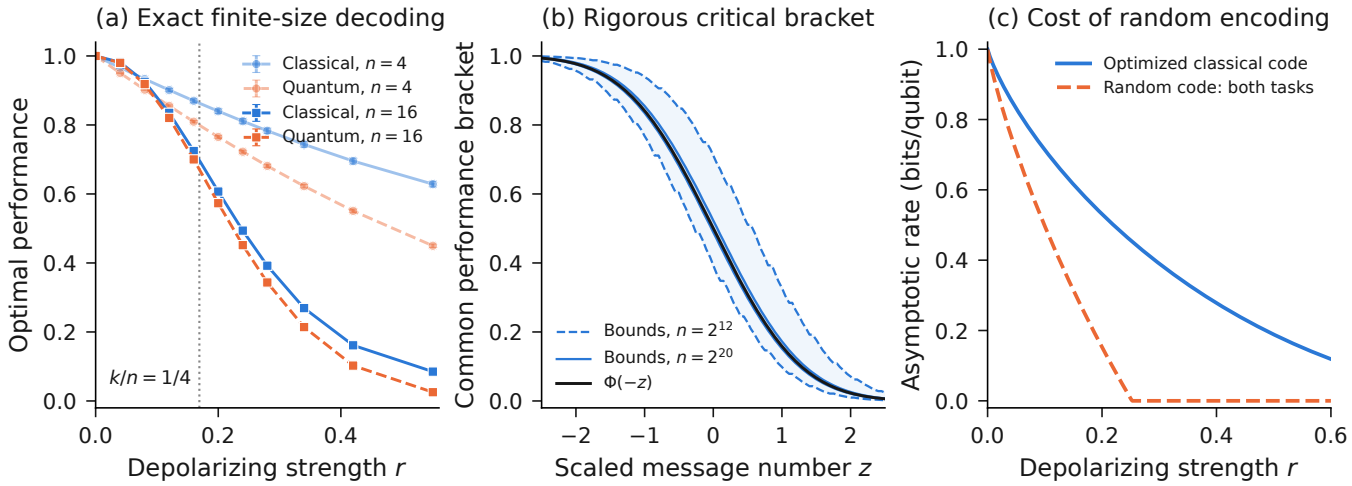


FIG. 1. Two optimal tasks and one asymptotic transition. (a) Exact classical success and entanglement fidelity for uniform Clifford encoders with $k/n = 1/4$; 64 encoders per size, reused across noise strengths. Markers show encoder means and one-standard-error bars; lines join points. The vertical line solves $H(1-3r/4, r/4, r/4, r/4) = 3/4$. The finite-size classical advantage is resolved. (b) Rigorous bounds in Eq. (5) for $r = 0.1$, $a = 2\log_2 n$, and k chosen by Eq. (15), compared with $\Phi(-z)$. These are binomial-CDF evaluations, not large-system decoding simulations. (c) The common random-code rate $[1-H]_+$ compared with the optimized classical capacity $1 - h_2(r/2)$ of the same depolarizing channel.

decay to zero for $R > \max(1-H, 0)$. More precisely, fix $\boldsymbol{p}$ with $0 < H < 1$ and $\sigma > 0$, and let

$$k_n = n(1-H) + \sqrt{n}\,\sigma z + O(1) \tag{15}$$

for fixed z. Berry–Esseen [13], applied to the two CDFs in Eq. (5) with $a = 2\log_2 n$, yields

$$\boxed{\begin{aligned} \mathbb{E}F_{\mathrm{q}}^* &= \Phi(-z) + O\left(\frac{\log n}{\sqrt{n}}\right), \\ \mathbb{E}P_{\mathrm{c}}^* &= \Phi(-z) + O\left(\frac{\log n}{\sqrt{n}}\right). \end{aligned}} \tag{16}$$

Here Φ is the normal CDF. Equation (6) with $\eta = n^{-1/2}$ shows that both quantities follow this curve for the same encoder outside an exceptional set of probability at most $37/(4n^{3/2}) + n^{5/2}/(4M_n)$. The constants are uniform over the choice of exact 2-design. At a fixed target error ϵ, the classical and quantum ensemble coding sizes therefore both obey

$$k_\epsilon = n(1-H) + \sqrt{n}\,\sigma\Phi^{-1}(\epsilon) + O(\log n). \tag{17}$$

Errors are respectively $1 - P_{\mathrm{c}}^*$ and $1 - F_{\mathrm{q}}^*$. Thus equality extends through the second-order term, not just the hashing rate. Fixed message dimension, zero surprisal variance, and the endpoint $H = 1$ are outside this Gaussian statement; Eq. (5) remains valid there.

Finite-size decoding.—We evaluate Eq. (4) for $n = 4, 8, 12, 16$ by an exact binary Fourier sum over syndrome and logical labels. Figure 1(a) shows how combining logical phase errors improves classical success at the smallest and largest sizes. Direct Pauli enumeration and primal–dual semidefinite optimization at $n \le 3$ independently check both optima and the explicit recovery. Methods, all sizes, seeds, and raw encoder values accompany the SM. The finite-size samples illustrate the decoder difference; the shared asymptotic curve is established by the spectral bounds [Fig. 1(b)].

Reliable decoding and retained information.—The classical task requires the entire message to be correct. A vanishing P_{c}^* can coexist with substantial partial information: a receiver who learns half of a uniform k-bit message and nothing about the rest has success $2^{-k/2}$, yet retains $k/2$ bits. Our converse bounds complete-message success for the specified random codebooks. It does not establish vanishing accessible information or failure to recover a smaller logical subsystem.

The distinction also separates this ensemble threshold from channel capacity. For $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, classical product-state coding achieves capacity $1 - h_2(r/2)$ [14]. At $r = 0.1$ it is 0.714 bits per qubit, whereas the common random-code rate is 0.497. The latter vanishes at $r_c = 0.252386\ldots$, where the classical capacity is still 0.453 [Fig. 1(c)]. Codes designed for classical messages therefore still support reliable communication where complete-message success for these random codebooks vanishes at every positive rate. Permitting the receiver to ignore coherence does not supply that encoding advantage.

Random-state contraction [15] and coherent information [6] probe related aspects of noisy encodings. Liu *et al.* also bound accessible information for a full state-2-design alphabet [16]. Here the alphabet is the M orthogonal columns at fixed, known U: the receiver optimizes separately for that alphabet before we average over U. A

classical converse for this conditional codebook matches quantum achievability, explaining the common recovery curve. Extending this result to approximate designs and noise accumulated during encoding would connect it to finite-depth devices.

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Data availability.—Source, raw encoder samples, numerical verification records, and figure data are supplied with this manuscript. No public archive identifier has yet been assigned.

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[10] See Supplemental Material for the full recovery proof, probability bounds, second-order inversion, Clifford optimality, and numerical methods.

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